Prayas JEE AIR (2025)

WEEKLY TEST 22

DURATION : 180 Minutes | **DATE : 21/12/2024** | **M.MARKS : 300**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25 questions** in each part in which first **20** questions are of Objective Type and Last **5 questions** are integers type and all **25 questions are compulsory**.
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number: _____

OMR Bar Code Number: _____

Candidate's Signature: _____ Invigilator's Signature _____

SECTION-I (PHYSICS)

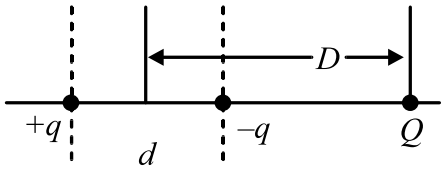
Single Correct Type Questions

1. In a car race on straight road, car *A* takes a time '*t*' less than car *B* at the finish and passes finishing point with a speed '*V*' more than that of car *B*. Both the cars start from rest and travel with constant acceleration a_1 and a_2 respectively. Then '*v*' is equal to

(1) $\frac{2a_1a_1}{a_1+a_2}t$ (2) $\sqrt{2a_1a_1}t$

(3) $\sqrt{a_1a_1}t$ (4) $\frac{a_1+a_2}{2}t$

2. A system of three charges are placed as shown in the figure If $D >> d$,the potential energy of the system is best given by



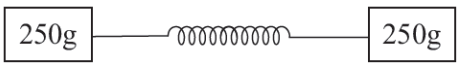
(1) $\frac{1}{4\pi\varepsilon_0}\left[-\frac{q^2}{d}-\frac{qQd}{D^2}\right]$

(2) $\frac{1}{4\pi\varepsilon_0}\left[+\frac{q^2}{d}+\frac{qQd}{D^2}\right]$

(3) $\frac{1}{4\pi\varepsilon_0}\left[-\frac{q^2}{d}+\frac{2qQd}{D^2}\right]$

(4) $\frac{1}{4\pi\varepsilon_0}\left[-\frac{q^2}{d}-\frac{qQd}{2D^2}\right]$

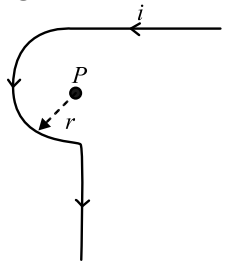
3. As per the given figure, two blocks each of mass 250g are connected to a spring of spring constant 2Nm^{-1}. If both are given velocity v = 0.5 m/sec. in opposite directions, then maximum elongation of the spring is:



(1) 50 cm (2) 25cm

(3) 75 cm (4) 80 cm

4. Find the magnetic field at the point *P* in figure. The curved portion is a semicircle connected to two long straight wires.



(1) $\frac{\mu_0 i}{2r}\left(1+\frac{2}{\pi}\right)$ (2) $\frac{\mu_0 i}{2r}\left(1+\frac{1}{\pi}\right)$

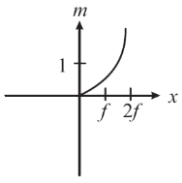
(3) $\frac{\mu_0 i}{2r}\left(\frac{1}{2}+\frac{1}{2\pi}\right)$ (4) $\frac{\mu_0 i}{2r}\left(\frac{1}{2}+\frac{1}{\pi}\right)$

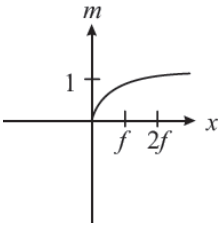
5. The torque of a force $5\hat{i}+3\hat{j}-7\hat{k}$ about the origin is τ. If the force acts on *a* particle whose position vector is $2\hat{i}+2\hat{j}+\hat{k}$, then the value of τ will be

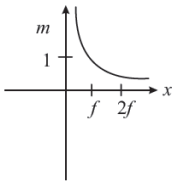
(1) $11\hat{i}+19\hat{j}+4\hat{k}$ (2) $-11\hat{i}+9\hat{j}-16\hat{k}$

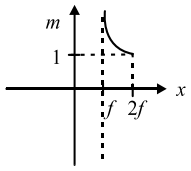
(3) $-17\hat{i}+19\hat{j}-4\hat{k}$ (4) $17\hat{i}+9\hat{j}+16\hat{k}$

6. An object is gradually moving away from the focal point of a concave mirror along the axis of the mirror. The graphical representation of the magnitude of linear magnification (*m*) versus distance of the object from the mirror (*x*) is correctly given by (Graphs are drawn schematically and are not to scale)

(1) 

(2) 

(3) 

(4) 

7. Young's moduli of the material of wires *A* and *B* are in the ratio of 1 : 4, while its area of cross sections are in the ratio of 1 : 3. If the same amount of load is applied to both the wires, the amount of elongation produced in the wires *A* and *B* will be in the ratio of [Assume length of wires *A* and *B* are same]

(1) 36 : 1 (2) 12 : 1

(3) 1 : 36 (4) 1 : 12

8. A linearly polarized electromagnetic wave in vacuum is

$$E=3.1\cos\left[(1.8)z-\left(5.4\times10^6\right)t\right]\hat{i}\ \text{N/C}$$

Is incident normally on a perfectly reflecting wall at $z = a$. Choose the correct option

(1) The frequency of electromagnetic wave is 54×10^4 Hz.

(2) The transmitted wave will be $3.1\cos\left[(1.8)z-\left(5.4\times10^6\right)t\right]\hat{i}\ \text{N/C}$

(3) The reflected wave will be $3.1\cos\left[(1.8)z+\left(5.4\times10^6\right)t\right]\hat{i}\ \text{N/C}$

(4) The wavelength is 5.4 m

9. Match the C_p/C_v ratio for ideal gases with different type of molecules

Molecule Type		C_p/C_v	
A.	Monoatomic	P	7/5
B.	Diatomic rigid molecules	Q	9/7
C.	Diatomic non-rigid molecules	R	4/3
D.	Triatomic rigid molecules	S	5/3

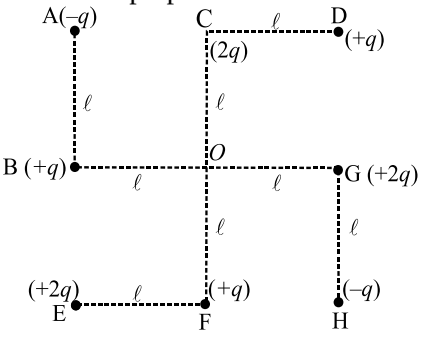
(1) A → Q; B → R; C → P; D → S
(2) A → S; B → Q; C → P; D → R
(3) A → S; B → P; C → Q; D → R
(4) A → R; B → S; C → Q; D → P

10. Consider the following fission chain reaction.

$$^{218}_{84}A \xrightarrow{\alpha} A_1 \xrightarrow{\beta^-} A_2 \xrightarrow{\gamma} A_3 \xrightarrow{\alpha} A_4 \xrightarrow{\beta^+} A_5 \xrightarrow{\gamma} A_6$$

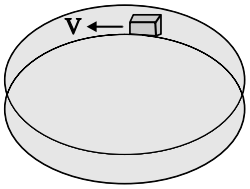
The mass number and the atomic number of A_6 are given by:
(1) 210 and 82
(2) 210 and 84
(3) 210 and 80
(4) 211 and 80

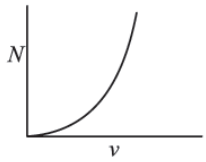
11. What will be the magnitude of electric field at point *O* as shown in figure? Each side of the figure is ℓ and perpendicular to each other?

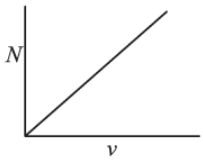


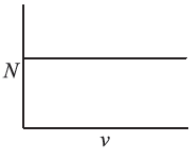
(1) $\frac{1}{4\pi\varepsilon_0}\frac{q}{(2\ell^2)}\left(2\sqrt{2}-1\right)$
(2) $\frac{q}{4\pi\varepsilon_0(2\ell)^2}$
(3) $\frac{1}{4\pi\varepsilon_0}\frac{q}{\ell^2}$
(4) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{2\ell^2}\left(\sqrt{2}\right)$

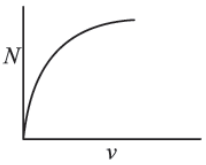
12. A smooth circular groove has a smooth vertical wall as shown in figure. A block of mass *m* moves against the wall with a speed *v*. Which of the following curve represents the correct relation between the normal reaction on the block by the wall (*N*) and speed of the block (*v*)?



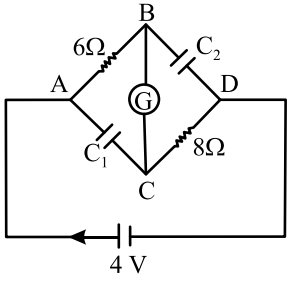
(1) 

(2) 

(3) 

(4) 

13. In this figure the resistance of the coil of galvanometer *G* is 2 Ω. The emf of the cell is 4 V. The ratio of potential difference across C_1 and C_2 is:

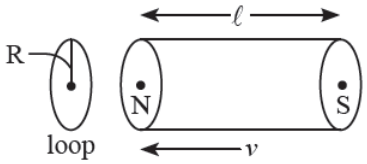


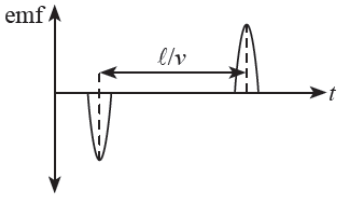
(1) 1 (2) $\frac{4}{5}$
(3) $\frac{3}{4}$ (4) $\frac{5}{4}$

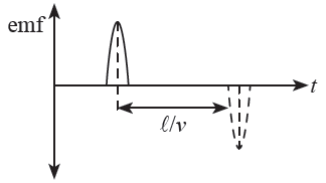
14. A rod of length *L* has non-uniform linear mass density given by $\rho(x) = a + b\left(\frac{x}{L}\right)^2$ where *a* and *b* are constants and $0 \le x \le L$. The value of *x* coordinate of the centre of mass of the rod

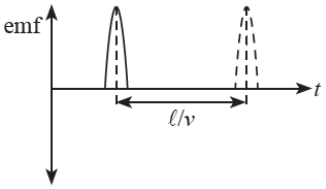
(1) $\frac{4}{3}\left(\frac{a+b}{2a+3b}\right)L$ (2) $\frac{3}{4}\left(\frac{2a+b}{3a+b}\right)L$
(3) $\frac{3}{2}\left(\frac{2a+b}{3a+b}\right)L$ (4) $\frac{3}{2}\left(\frac{a+b}{2a+b}\right)L$

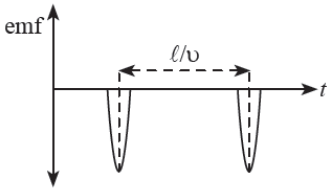
15. A bar magnet is passing through a conducting loop of radius R with velocity v. The radius of the bar magnet is such that it just passes through the loop. The induced e.m.f. in the loop can be represented by the approximate curve:



(1)


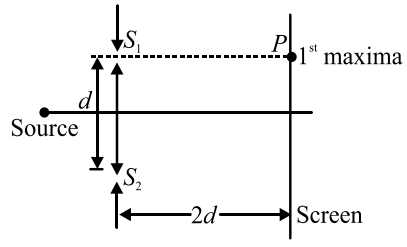
(2)


(3)


(4)


16. Given below are two statements.
Statement-I: The law of gravitation holds good for any pair of bodies in the universe.
Statement-II: The weight of any person becomes zero when the person is at the centre of the earth.
In the light of the above statements, choose the correct answer from the option given below.
(1) Both Statement-I and Statement-II are true
(2) Both Statement-I and Statement-II are false
(3) Statement-I is true but Statement-II is false
(4) Statement-I is false but Statement-II is true

17. Consider a Young's double slit experiment as shown in figure. What should be the slit separation d in terms of wavelength λ such that the first maxima. occurs directly in front of the slit (S_1)?



(1) $\frac{\lambda}{2\left(\sqrt{5}-2\right)}$ (2) $\frac{\lambda}{\left(\sqrt{5}-2\right)}$

(3) $\frac{\lambda}{2\left(5-\sqrt{2}\right)}$ (4) $\frac{\lambda}{\left(5-\sqrt{2}\right)}$

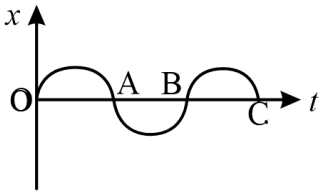
18. Consider a solid sphere of radius R and mass density $\rho(r)=\rho_0\left(1-\frac{r^2}{R^2}\right), 0<r\le R$. The minimum density of a liquid in which it will float is

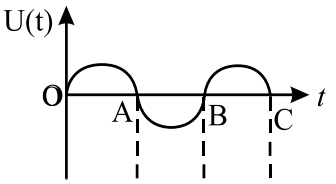
(1) $\frac{\rho_0}{5}$ (2) $\frac{2\rho_0}{5}$

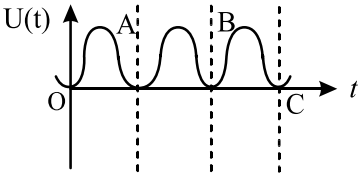
(3) $\frac{2\rho_0}{3}$ (4) $\frac{\rho_0}{3}$

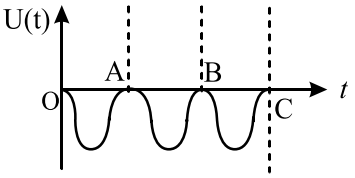
19. An electron of a hydrogen like atom, having Z = 4, jumps from 4th energy state to 2nd energy state, The energy released in this process, will be:
(Given Rch = 13.6 eV)
Where R = Rydberg constant
c = Speed of light in vacuum
h = Planck's constant
(1) 13.6 eV (2) 10.5 eV
(3) 3.4 eV (4) 40.8 eV

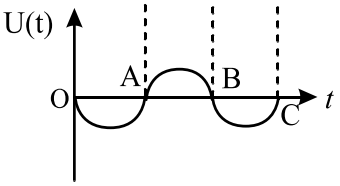
20. The variation of displacement with time of a particle executing free simple harmonic motion is shown in the figure. The potential energy U(t) versus time (t) plot of the particle is correctly shown in figure:



(1)


(2)


(3)


(4)


Integer Type Questions

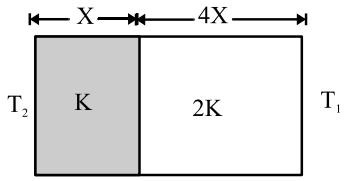
21. An organ pipe P_1 closed at one end vibrating in its first overtone and another pipe P_2 open at both ends vibrating in third overtone are in resonance with a given tuning fork. If the ratio of the length of P_1 to that of P_2 is ϕ than find the value of 8ϕ .

22. A parallel plate capacitor has 1μF capacitance. One of its two plates is given +2μC charge and the other plate, +4μC charge. The potential difference (in volt) developed across the capacitor is :

23. An alternating voltage V(t) = 220 sin $\left(\frac{50}{3}\pi t\right)$ volt here t is in seconds is applied to a purely resistive load of 50 Ω The time taken (in ms) for the current to rise from half of the peak value to the peak value is:

24. A metal plate of area 1×10^{-4} m^2 is illuminated by a radiation of intensity 16 mW/m^2. The work function of the metal is 5 eV. The energy of the incident photons is 10 eV and only 10% of it produces photo electrons. The number of emitted photo electrons per second is $(10)^x$ than find the value of x ?
[1 eV = 1.6×10^{-19} J]

25. The temperature of the two outer surfaces of a composite slab, consisting of two materials having coefficient of thermal conductivity K and $2K$ and thickness x and $4x$ respectively are T_2 and T_1 ($T_2 > T_1$). The rate of heat transfer through the slab, in a steady state is $\left(\frac{A(T_2 - T_1)K}{x}\right) f$ than the value of $\frac{1}{f}$.

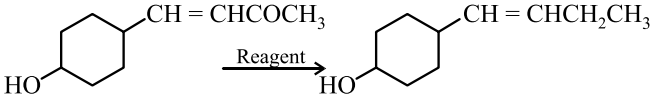


SECTION-II (CHEMISTRY)

Single correct Type Questions

26. The energy required to break one mole of Cl–Cl bonds in Cl_2 is 242 kJ mol^{-1}. The longest wavelength of light capable of breaking a single CI–Cl bond is (c = 3×10^8 ms^{-1} and N_A = 6.02×10^{23} mol^{-1})
(1) 594 nm
(2) 640 nm
(3) 700 nm
(4) 494 nm

27. In the given transformation, which of the following is the most appropriate reagent?



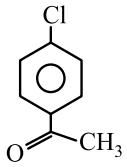
(1) NH_2NH_2, OH^-
(2) Zn-Hg/HCl
(3) Na.Liq.NH_3
(4) $NaBH_4$

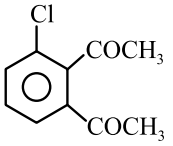
28. The value of K_C is 64 at 800 K for the reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
The value of K_C for the following reaction is:

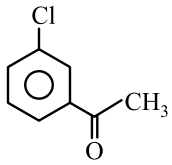
$$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$$

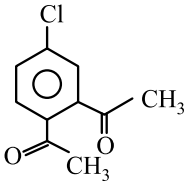
(1) $\frac{1}{64}$ (2) 8
(3) $\frac{1}{4}$ (4) $\frac{1}{8}$

29. The major product formed in the Friedel-Craft acylation of chlorobenzene is.

(1) 

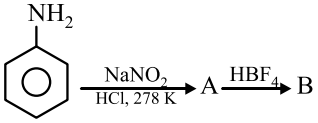
(2) 

(3) 

(4) 

30. Identify the process in which change in the oxidation state is five:
(1) $CrO_4^{2-} \rightarrow Cr^{3+}$ (2) $MnO_4^- \rightarrow Mn^{2+}$
(3) $Cr_2O_7^{2-} \rightarrow 2Cr^{3+}$ (4) $C_2O_4^{2-} \rightarrow 2CO_2$

31. In the chemical reactions,



the compounds 'A' and 'B' respectively are
(1) nitrobenzene and fluorobenzene
(2) phenol and benzene
(3) benzene diazonium chloride and fluorobenzene
(4) nitrobenzene and chlorobenzene

32. Given: $E^{\circ}_{Fe^{3+}/Fe} = -0.036\,V, E^{\circ}_{Fe^{2+}/Fe} = -0.439\,V$. The value of standard electrode potential for the change, $Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$ will be :

(1) –0.072 V (2) 0.385 V
(3) 0.770 V (4) –0.270 V

33. A certain orbital has no angular nodes and two radial nodes. The orbital is:

(1) 3s
(2) 3p
(3) 2s
(4) 2p

34. The oxidation states of nitrogen in NO, NO_2, N_2O and NO_3^- are in the order of:

(1) $NO_2 > NO_3^- > NO > N_2O$
(2) $NO_3^- > NO_2 > NO > N_2O$
(3) $NO > NO_2 > N_2O > NO_3^-$
(4) $N_2O > NO_2 > NO > NO_3^-$

35. Match the **Column**-I with **Column**-II.

Column-I Test / Method		**Column-II Reagent**	
A	Lucas Test	**P**	$C_6H_5SO_2Cl$/aq.kOH
B	Dumas method	**Q**	Conc. H_2SO_4
C	Kjeldahl's method	**R**	CuO/CO_2
D	Hinsberg Test	**S**	Conc. HCl and $ZnCl_2$

(1) A→R; B→S; C→Q; D→P
(2) A→Q; B→S; C→R; D→P
(3) A→S; B→R; C→Q; D→P
(4) A→Q; B→P; C→R; D→S

36. What is the correct order of the following elements with respect to their density?

(1) Zn < Cr < Fe < Co < Cu
(2) Zn < Cu < Co < Fe < Cr
(3) Cr < Fe < Co < Cu < Zn
(4) Cr < Zn < Co < Cu < Fe

37. The correct shape and I – I – I bond angles respectively in I_3^- ion are:

(1) Linear; 180°
(2) T-shaped ; 180° and 90°
(3) Trigonal planar; 120°
(4) Distorted trigonal planar; 135° and 90°

38. $H_3C–CH_2–\underset{\underset{OH}{|}}{CH}–CH_3 \xrightarrow[(iii)D_2O]{\substack{(i)NaI,\ H_3PO_4\\(ii)Mg,\ Dry\ ether}} [X]$ Product

Product [X] formed in the above reaction is:

(1) $H_3C–CH_2–\underset{\underset{D}{|}}{CH}–CH_3$

(2) $H_3C–CH_2–\underset{\underset{OH}{|}}{\overset{\overset{H}{|}}{C}}–CH_3$

(3) $H_3C—CH_2—CH=CH_2$
(4) $H_3C—CH=CH—CH_3$

39. B has a smaller first ionization enthalpy than Be. Consider the following statements:

(I) it is easier to remove 2p electron than 2s electron.
(II) 2p electron of B is more shielded from the nucleus by the inner core of electrons than the 2s electrons of Be.
(III) 2s electron has more penetration power than 2p electron
(IV) atomic radius of B is more than Be (atomic number B = 5, Be = 4)

The correct statements are:

(1) (I), (III) and (IV)
(2) (I), (II) and (III)
(3) (I), (II) and (IV)
(4) (II), (III) and (IV)

40. One mole of Compound A gives two moles of D-Glucose on hydrolysis. The compound A is:

(1) Amylose
(2) Sucrose
(3) Maltose
(4) Lactose

41. Given below are two statements:

Statement-I: The limiting molar conductivity of KCl (strong electrolyte) is higher compared to that of CH_3COOH (weak electrolyte).

Statement-II: Molar conductivity decreases with decrease in concentration of electrolyte.

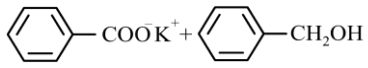
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement-I is false but Statement-II is true.
(2) Both Statement-I and Statement-II are false.
(3) Both Statement-I and Statement-II are true.
(4) Statement-I is true but Statement-II is false.

42. The compound A in the following reactions is:

(A) $\xrightarrow[(ii)Conc.H_2SO_4/\Delta]{(i)CH_3MgBr/H_2O}$

(B) $\xrightarrow[(ii)Zn/H_2O]{(i)O_3}$ C + D

(C) $\xrightarrow[(ii)\Delta]{(i)Conc.\ KOH}$ 

(D) $\xrightarrow[\Delta]{Ba(OH)_2}$ $H_3C-\underset{\underset{CH_3}{|}}{C}=CH-\underset{\underset{O}{\|}}{C}-CH_3$

(1) $C_3H_5-\underset{\underset{O}{\|}}{C}-CH_2CH_3$

(2) $C_6H_5-CH_2-\underset{\underset{O}{\|}}{C}-CH_3$

(3) $C_6H_5-\underset{\underset{O}{\|}}{C}-CH_3$

(4) $C_6H_5-\underset{\underset{O}{\|}}{C}-CH\begin{matrix}CH_3\\CH_3\end{matrix}$

43. The reagent from those given below, that can not convert nitrobenzene into aniline is ____. (Integer answer)

(1) Sn–HCl (2) Sn–NH_4OH

(3) Fe–HCl (4) Zn–HCl

44. For the equilibrium

$2H_2O \rightleftharpoons H_3O^+ + OH^-$, the value of $\Delta G°$ at 298 K is approximately

(1) 100 kJ mol^{-1} (2) –80 kJ mol^{-1}

(3) 80 kJ mol^{-1} (4) –100 kJ mol^{-1}

45. The metal d-orbital that can directly facing the ligands in $K_3[Co(CN)_6]$ are

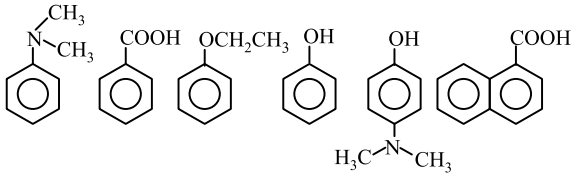
(1) d_{xy} and $d_{x^2-y^2}$ (2) $d_{x^2-y^2}$ and d_{z^2}

(3) d_{xz}, d_{yz} and d_{z^2} (4) d_{xy}, d_{xz} and d_{yz}

Integer type Questions

46. A one molal aqueous $K_4Fe(CN)_6$ solution has a degree of dissociation of 0.4. Its boiling point is equal to that of another aqueous solution which contains 18.1 weight percent of a non electrolytic solute A. The molar mass of A is ____g/mol. (Round off to the Nearest Integer). [Density of water = 1.0 g cm^{-3}].

47. Amongst the following the total number of compounds soluble in aqueous NaOH is



48. Number of bromo derivatives obtained on treating ethane with excess of Br_2, in diffused sunlight is _____.

49. $H_3C-\underset{\underset{CH_3}{|}}{\overset{\overset{O}{/\backslash}}{C}}-CH_2 \xrightarrow[(ii)\ H_2O/H^+]{(i)\ CH_3MgBr} (P) \xrightarrow[\Delta]{H^+} (Q)$

How many hyperconjugating structures of "Q" are possible?

50. $\underset{\text{Reactant}}{[A]} \rightarrow \underset{\text{Product}}{[B]}$

If formation of compound [B] follows the first order of kinetics and after 70 minutes the concentration of [A] was found to be half of its initial concentration. Then the rate constant of the reaction is x $\times 10^{-6}$ s^{-1}. The value of x is_____. (use, ln2 = 0.693)

SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. Let α be a root of equation $x^2 + x + 1 = 0$ and the matrix $A = \frac{1}{\sqrt{3}}\begin{bmatrix}1 & 1 & 1\\1 & \alpha & \alpha^2\\1 & \alpha^2 & \alpha^4\end{bmatrix}$ then the matrix A^{31} is equal to

(1) A^3 (2) A^2

(3) I_3 (4) A

52. Let A, B, C be 3 × 3 matrices such that A is symmetric and B and C are skew-symmetric. Consider the statements

(S1) $A^{13}B^{26}-B^{26}A^{13}$ is symmetric

(S2) $A^{26}C^{13}-C^{13}A^{26}$ is symmetric

Then,

(1) Only S2 is true

(2) Only S1 is true

(3) Both S1 and S2 are false

(4) Both S1 and S2 are true

53. The natural number m, for which the coefficient of x in the binomial expansion of $\left(x^m + \frac{1}{x^2}\right)^{22}$ is 1540, is.

(1) 3

(2) 4

(3) 12

(4) 13

54. If $a>0$ and $z=\frac{(1+i)^2}{a-i}$ has magnitude $\sqrt{\frac{2}{5}}$ then $\bar{z}$ is equals to

(1) $-\frac{3}{5}-\frac{1}{5}i$ (2) $-\frac{1}{5}-\frac{3}{5}i$

(3) $-\frac{1}{5}+\frac{3}{5}i$ (4) $\frac{1}{5}-\frac{3}{5}i$

55. Let the point $P(\alpha, \beta)$ be at a unit distance from each of the two lines $L_1 : 3x - 4y + 12 = 0$, and $L_2 : 8x + 6y + 11 = 0$. If P lies below L_1 and above L_2, then $100(\alpha + \beta)$ is equal to

(1) –14
(2) 42
(3) –22
(4) 14

56. The shortest distance between the lines $x + 1 = 2y = -12z$ and $x = y + 2 = 6z - 6$ is

(1) 2 (2) 3

(3) $\frac{5}{2}$ (4) $\frac{3}{2}$

57. Consider the following frequency distribution:

Class	0 – 6	6 – 12	12 – 18	18 – 24	24 – 30
Frequency	*a*	*b*	12	9	5

If mean $=\frac{309}{22}$ and median =14, then the value $(a-b)^2$ is equal to

(1) 1 (2) 2
(3) 3 (4) 4

58. Let S_n denote the sum of the first n terms of an A.P. If $S_4 = 16$ and $S_6 = -48$, then S_{10} is equal to:

(1) –320 (2) –260
(3) –380 (4) –410

59. If the x-intercept of a focal chord of the parabola $y^2 = 8x + 4y + 4$ is 3, then the length of this chord is equal to _____.

(1) 17 (2) 14
(3) 16 (4) 18

60. Let f be a differentiable function in $\left(0,\frac{\pi}{12}\right)$. If $\int_{\cos x}^{1} t^2 f(t)dt = \sin^3 x + \cos x$ then $\frac{1}{\sqrt{3}} f'\left(\frac{1}{\sqrt{3}}\right)$ is equal to

(1) $6-9\sqrt{2}$ (2) $6-\frac{9}{\sqrt{2}}$

(3) $\frac{9}{2}-6\sqrt{2}$ (4) $\frac{9}{\sqrt{2}}-6$

61. The number of five digit numbers, greater than 40000 and divisible by 5, which can be formed using the digits 0, 1, 3, 5, 7 and 9 without repetition, is equal to

(1) 120
(2) 132
(3) 72
(4) 96

62. The set of values of k, for which the circle $C: 4x^2 + 4y^2 - 12x + 8y + k = 0$ lies inside the fourth quadrant and the point $\left(1,-\frac{1}{3}\right)$ lie on or inside the circle C, is.

(1) An empty set (2) $\left(6,\frac{65}{9}\right]$

(3) $\left[\frac{80}{9},10\right)$ (4) $\left(9,\frac{92}{9}\right]$

63. The probability of selecting integers $a \in [-5, 30]$ such that $x^2 + 2(a + 4)x - 5a + 64 > 0$, for all $x \in R$ is:

(1) $\frac{2}{9}$ (2) $\frac{1}{6}$

(3) $\frac{7}{36}$ (4) $\frac{1}{4}$

64. Consider ellipses $E_k : kx^2 + k^2 y^2 = 1$, $k = 1, 2, \ldots, 20$. Let C_k be the circle which touches the four chords joining the end points (one on minor axis and another on major axis) of the ellipse E_k, If r_k is the radius of the circle C_k, then the value of $\sum_{k=1}^{20}\frac{1}{r_k^2}$ is

(1) 3080 (2) 3210
(3) 3320 (4) 2870

65. Let $\vec{a}$ be a non-zero vector parallel to the line of intersection of the two planes described by $\hat{i}+\hat{j}, \hat{i}+\hat{k}$ and $\hat{i}-\hat{j}, \hat{j}-\hat{k}$. If θ is the angle between the vector $\vec{a}$ and the vector $\vec{b}=2\hat{i}-2\hat{j}+\hat{k}$ and $\vec{a}\cdot\vec{b}=6$ then the ordered pair $(\theta, |\vec{a}\times\vec{b}|)$ is equal to

(1) $\left(\frac{\pi}{4},3\sqrt{6}\right)$ (2) $\left(\frac{\pi}{3},3\sqrt{6}\right)$

(3) $\left(\frac{\pi}{3},6\right)$ (4) $\left(\frac{\pi}{4},6\right)$

66. Let $lx - 2y = 1$ intersects the parabola $y^2 = 4x$ at points P and Q. If PS and QS meet the parabola again at R and T respectively (where S is the focus of the parabola), then the slope of RT is equal to

(1) 5 (2) 2
(3) 0.5 (4) 0.2

67. Let $x = x(y)$ be the solution of the differential equation $2(y + 2)\log_e (y + 2)dx + (x + 4 - 2\log_e (y + 2))dy = 0$, $y > -1$ with $x(e^4 - 2) = 1$. Then $x(e^9 - 2)$ is equal to

(1) $\frac{4}{9}$ (2) $\frac{10}{3}$

(3) $\frac{32}{9}$ (4) 3

68. The area of the region enclosed by $y \le 4x^2, x^2 \le 9y$ and $y \le 4$ is equal to

(1) $\frac{40}{3}$ (2) $\frac{56}{3}$

(3) $\frac{112}{3}$ (4) $\frac{80}{3}$

69. The function, $f(x) = (3x - 7)x^{23}, x \in R$, is increasing for all x lying in:

(1) $(-\infty, 0) \cup \left(\frac{14}{15}, \infty\right)$

(2) $(-\infty, 0) \cup \left(\frac{3}{7}, \infty\right)$

(3) $\left(-\infty, -\frac{14}{15}\right) \cup (0, \infty)$

(4) $\left(-\infty, \frac{14}{15}\right)$

70. The remainder when $7^{2022} + 3^{2022}$ is divided by 5 is:

(1) 0 (2) 2

(3) 3 (4) 4

Integer Type Questions

71. Let $\alpha > 0$. If $\int_0^{\alpha} \frac{x}{\sqrt{x+\alpha} - \sqrt{x}} dx = \frac{16 + 20\sqrt{2}}{15}$, then α is equal to:

72. Considering only the principal values of the inverse trigonometric functions, the value of $\frac{3}{2}\cos^{-1}\sqrt{\frac{2}{2+\pi^2}} + \frac{1}{4}\sin^{-1}\frac{2\sqrt{2}\pi}{2+\pi^2} + \tan^{-1}\frac{\sqrt{2}}{\pi}$ is $\frac{k\pi}{4}$, then k is

73. Let $I(x) = \int \sqrt{\frac{x+7}{x}} dx$ and $I(9) = 12 + 7\log_e 7$. If $I(1) = \alpha + 7\log_e (1 + 2\sqrt{2})$, then α^4 is

74. Let $A = \{1, 2, 3, 4\}$ and R be a relation on the set $A \times A$ defined by $R = \{((a, b), (c, d)): 2a + 3b = 4c + 5d\}$. Then the number of elements in R is:

75. $16\sin(20°)\sin(40°)\sin(80°)$ is equal to $2\sqrt{k}$, then k is



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